

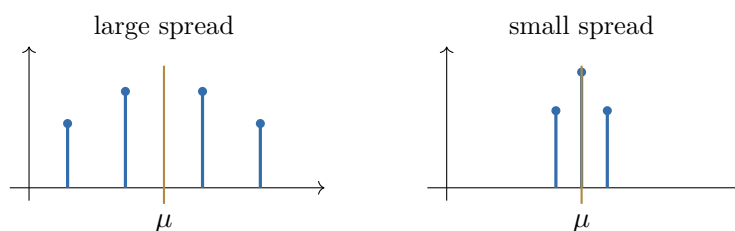
Chapter 10

Variance; Independent Trials; Binomial, Geometric and Hypergeometric

Lecture 10 | Devore 9e: 3.3–3.5

10.1 Variance

The mean locates the distribution. It says nothing about how spread out it is: the two pmfs below have the same μ and could hardly be more different.



The natural measure of spread is the average squared distance from the mean.

Definition 10.1 (Variance and standard deviation). Let X have mean $\mu = \mathbb{E}[X]$. The **variance** is

$$V(X) = \text{Var}(X) = \sigma_X^2 := \mathbb{E}[(X - \mu)^2],$$

and the **standard deviation** is $\sigma_X = \sqrt{\text{Var}(X)}$.

Note that $\text{Var}(X)$ is just $\mathbb{E}[h(X)]$ for $h(x) = (x - \mu)^2$, so the previous lecture already tells us how to compute it: $\text{Var}(X) = \sum_{x_i \in D} (x_i - \mu)^2 \mathbb{P}\{X = x_i\}$.

Remark 10.2. Why square rather than take absolute values? Squaring is smooth (differentiable), it makes the algebra below work, and it is what turns independence into additivity later (Lecture 20). The price is that σ^2 has the *squared* units of X — which is why we also report σ , which is in the original units.

Theorem 10.3 (Computational formula).

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Proof. Expand and use linearity, remembering μ is a constant:

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2. \quad \square$$

So computing a variance is always the same three steps:

$$(1) \mathbb{E}[X], \quad (2) \mathbb{E}[X^2], \quad (3) \text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

Example 10.4 (Both routes on the same numbers). A loaded die has pmf

x	1	2	3	4	5	6
$p(x)$	0.30	0.25	0.15	0.05	0.10	0.15

(The probabilities sum to 1. ✓) First the mean:

$$\mathbb{E}[X] = 1(0.30) + 2(0.25) + 3(0.15) + 4(0.05) + 5(0.10) + 6(0.15) = 0.30 + 0.50 + 0.45 + 0.20 + 0.50 + 0.90 = 2.85.$$

Route 1 — straight from the definition.

$$\begin{aligned} \text{Var}(X) &= \sum_{x=1}^6 (x - 2.85)^2 p(x) \\ &= (-1.85)^2(0.30) + (-0.85)^2(0.25) + (0.15)^2(0.15) \\ &\quad + (1.15)^2(0.05) + (2.15)^2(0.10) + (3.15)^2(0.15) \\ &= 1.026750 + 0.180625 + 0.003375 + 0.066125 + 0.462250 + 1.488375 \\ &= 3.2275. \end{aligned}$$

Route 2 — the computational formula.

$$\mathbb{E}[X^2] = 1(0.30) + 4(0.25) + 9(0.15) + 16(0.05) + 25(0.10) + 36(0.15) = 0.30 + 1.00 + 1.35 + 0.80 + 2.50 + 5.40 = 11.35,$$

$$\text{Var}(X) = 11.35 - (2.85)^2 = 11.35 - 8.1225 = 3.2275. \quad \checkmark$$

Either way $\sigma = \sqrt{3.2275} \approx 1.80$.

Remark 10.5. Route 2 is almost always less work — six squarings of small integers instead of six squarings of awkward decimals — and it is far less error-prone. Use the definition to understand what variance *means*; use $\mathbb{E}[X^2] - \mu^2$ to compute it.

Proposition 10.6. $\text{Var}(aX + b) = a^2 \text{Var}(X)$ and $\sigma_{aX+b} = |a| \sigma_X$.

Proof. $\mathbb{E}[aX + b] = a\mu + b$, so $(aX + b) - (a\mu + b) = a(X - \mu)$, and squaring gives $a^2(X - \mu)^2$. Taking expectations, $\text{Var}(aX + b) = a^2 \mathbb{E}[(X - \mu)^2] = a^2 \text{Var}(X)$; the standard deviation is the square root, so it picks up $|a|$ rather than a . $\square$

Shifting does not change spread; rescaling scales it. That is exactly what a measure of spread should do.

Example 10.7 (Variance of the shooting variable). With $\mathbb{P}\{X = n\} = (1-\beta)^{n-1}\beta$ and $\mathbb{E}[X] = 1/\beta$, we need $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = \beta \sum_{n=1}^{\infty} n^2 (1-\beta)^{n-1}.$$

Differentiate the geometric series twice. Since $\frac{d^2}{d\beta^2}(1-\beta)^{n+1} = (n+1)n(1-\beta)^{n-1} = n^2(1-\beta)^{n-1} + n(1-\beta)^{n-1}$, summing over $n \geq 1$ and differentiating term by term reduces everything to one series we can add up in closed form:

$$\sum_{n \geq 1} (1-\beta)^{n+1} = \frac{(1-\beta)^2}{\beta} = \frac{1}{\beta} - 2 + \beta, \quad \frac{d^2}{d\beta^2} \left(\frac{1}{\beta} - 2 + \beta \right) = \frac{2}{\beta^3},$$

so $\sum_{n \geq 1} (n^2 + n)(1-\beta)^{n-1} = \frac{2}{\beta^3}$. The first-derivative result of Lecture 9 already gave $\sum_{n \geq 1} n(1-\beta)^{n-1} = \frac{1}{\beta^2}$, and subtracting it leaves $\sum_{n \geq 1} n^2(1-\beta)^{n-1} = \frac{2}{\beta^3} - \frac{1}{\beta^2}$. Putting the pieces together,

$$\mathbb{E}[X^2] = \beta \left(\frac{2}{\beta^3} - \frac{1}{\beta^2} \right) = \frac{2}{\beta^2} - \frac{1}{\beta} = \frac{2-\beta}{\beta^2}, \quad \text{hence} \quad \text{Var}(X) = \frac{2-\beta}{\beta^2} - \frac{1}{\beta^2} = \boxed{\frac{1}{\beta^2} - \frac{1}{\beta} = \frac{1-\beta}{\beta^2}}.$$

10.2 Independent trials

The binomial and geometric distributions below both come from repeating one experiment independently, and the hypergeometric at the end is what happens when the repetitions are *not* independent. We have multiplied the probabilities of independent events since Lecture 5, and the shooting variable of Lecture 8 already assumed independent shots; now we make “independent repetitions” precise.

So far independence has been a property of events inside one experiment. Usually we want it to be a property of *experiments*: repeated coin tosses, repeated measurements, repeated draws. That requires building a new probability space out of two old ones.

Let Experiment 1 have space $(S_1, \mathcal{F}_1, \mathbb{P}_1)$ and Experiment 2 have $(S_2, \mathcal{F}_2, \mathbb{P}_2)$. Take the **product**

$$S := S_1 \times S_2, \quad \mathcal{F} := \mathcal{F}_1 \times \mathcal{F}_2,$$

and define $\mathbb{P}$ on rectangles $A = A_1 \times A_2$ by

$$\mathbb{P}(A_1 \times A_2) := \mathbb{P}_1(A_1) \cdot \mathbb{P}_2(A_2).$$

Definition 10.8 (Independent trials). With the above $\mathbb{P}$, the triple $(S, \mathcal{F}, \mathbb{P})$ is a probability space, and Experiments 1 and 2 are called **independent trials**.

Example 10.9. $S_1 = \{1, 2, 3, 4, 5, 6\}$ (a die), $S_2 = \{\text{red}, \text{black}\}$ (a card color). Then $S_1 \times S_2 = \{(1, \text{red}), (1, \text{black}), (2, \text{red}), \dots\}$, with twelve outcomes, and $\mathbb{P}((3, \text{red})) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}$.

Example 10.10 (Calm gambler). One game: $\mathbb{P}(W) = 0.6$, $\mathbb{P}(L) = 0.4$, and games are independent trials. Play three games, so $S = \{W, L\}^3$. Then

$$\mathbb{P}(WWW) = \mathbb{P}(W)\mathbb{P}(W | W)\mathbb{P}(W | WW) = \mathbb{P}(W)^3 = 0.6^3 = 0.216,$$

$$\mathbb{P}(LLW) = 0.4 \times 0.4 \times 0.6 = 0.096.$$

Independence is exactly what lets us drop the conditioning bars.

Example 10.11 (Emotional gambler). The same chances in the *first* game, $\mathbb{P}(W) = 0.6$, $\mathbb{P}(L) = 0.4$, but the player is rattled by wins and motivated by losses:

$$\mathbb{P}(W | W) = 0.2, \quad \mathbb{P}(L | W) = 0.8, \quad \mathbb{P}(W | L) = 0.8, \quad \mathbb{P}(L | L) = 0.2,$$

and the player reacts only to the game just played, so $\mathbb{P}(W | WW) = \mathbb{P}(W | W)$ and so on. These are *not* independent trials, so we must keep the multiplication rule in full:

$$\mathbb{P}(WWW) = 0.6 \times 0.2 \times 0.2 = 0.024, \quad \mathbb{P(LLW)} = 0.4 \times 0.2 \times 0.8 = 0.064.$$

Compare with the calm gambler: same first-game odds, very different three-game behavior.

10.3 The binomial distribution

From here on we stop building distributions by hand and start naming the ones that occur over and over.

Definition 10.12 (Binomial experiment). An experiment consisting of n *independent* trials, where

- each trial has exactly two outcomes, success (S) and failure (F);
- $\mathbb{P}(S) = p$ and $\mathbb{P}(F) = 1 - p \stackrel{\text{def}}{=} q$, the same on every trial.

Definition 10.13 (Binomial distribution). In a binomial experiment let $X :=$ the number of successes among the n trials. Then $X \sim \text{Binomial}(n, p)$, and its pmf is written $b(x; n, p)$.

Theorem 10.14.

$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. The sample space consists of all strings of length n over $\{S, F\}$, e.g. $SSF \cdots FS$. By independence, any *particular* string with x successes has probability

$$\mathbb{P}(S)\mathbb{P}(S)\mathbb{P}(F) \cdots \mathbb{P}(F)\mathbb{P}(S) = p^x (1-p)^{n-x},$$

regardless of the order the letters appear in. The event $\{X = x\}$ is the disjoint union of all such strings, and there are $\binom{n}{x}$ of them (choose which x of the n positions hold the successes). Adding up, $\mathbb{P}\{X = x\} = \binom{n}{x} p^x (1-p)^{n-x}$. $\square$

Theorem 10.15. If $X \sim \text{Binomial}(n, p)$ then $\mathbb{E}[X] = np$ and $\text{Var}(X) = np(1-p)$.

(The slick proof writes $X = \mathbf{1}_1 + \cdots + \mathbf{1}_n$ as a sum of indicators; we give it in Lecture 20 once we have additivity of variance.)

Remark 10.16. Half of that slick proof is already available. By linearity of expectation (Lecture 9), $\mathbb{E}[X] = \mathbb{E}[\mathbf{1}_1] + \cdots + \mathbb{E}[\mathbf{1}_n] = p + \cdots + p = np$, with no series at all. It is the variance that has to wait, because variance is not linear.

10.4 The geometric distribution

Definition 10.17 (Geometric distribution). In a binomial experiment, let $X :=$ the number of trials up to and including the first success. Then $X \sim \text{Geometric}(p)$ with pmf

$$g(x; p) = (1 - p)^{x-1}p, \quad x = 1, 2, 3, \dots$$

This is the shooting-at-a-target variable from Lectures 8–9, so we already know

$$\mathbb{E}[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1}{p^2} - \frac{1}{p} = \frac{1-p}{p^2}.$$

Remark 10.18. Binomial and geometric come from the *same* experiment; they differ in what is fixed and what is random. Binomial: n fixed, count the successes. Geometric: one success required, count the trials. It is named after the geometric series, which is where its normalization and its mean came from.

Remark 10.19 (Warning). Some books (and software) define the geometric as the number of *failures* before the first success, i.e. shifted down by one, with mean $(1-p)/p$. We always use the “number of trials” convention, starting at $x = 1$.

Theorem 10.20 (Tail-sum formula). *If N takes values in $\{0, 1, 2, \dots\}$, then*

$$\mathbb{E}[N] = \sum_{k=1}^{\infty} \mathbb{P}\{N \geq k\}.$$

Proof. Write $\mathbb{E}[N] = \sum_{n \geq 1} n \mathbb{P}\{N = n\} = \sum_{n \geq 1} \sum_{k=1}^n \mathbb{P}\{N = n\}$ and exchange the order of summation, which is allowed because every term is non-negative:

$$\sum_{k \geq 1} \sum_{n \geq k} \mathbb{P}\{N = n\} = \sum_{k \geq 1} \mathbb{P}\{N \geq k\}. \quad \square$$

Tail probabilities are often much easier than point probabilities. For $N \sim \text{Geometric}(p)$ the event $\{N \geq k\}$ just says that the first $k-1$ trials failed, so

$$\mathbb{E}[N] = \sum_{k \geq 1} (1-p)^{k-1} = \frac{1}{p}$$

— the answer of Lecture 9, with no derivative trick.

Example 10.21 (Collecting all six faces). A die is rolled until every face has appeared at least once. How many rolls does that take on average?

Split the wait into stages. Let G_j be the number of rolls needed to see a new face once $j-1$ different faces have already appeared, so the total is $N = G_1 + \dots + G_6$. During stage j each roll shows a new face with probability $\frac{7-j}{6}$, independently of the past, so $G_j \sim \text{Geometric}(\frac{7-j}{6})$ with mean $\frac{6}{7-j}$. By linearity (Lecture 9),

$$\mathbb{E}[N] = 6 \left(\frac{1}{6} + \frac{1}{5} + \frac{1}{4} + \frac{1}{3} + \frac{1}{2} + 1 \right) = \boxed{14.7}.$$

The last face alone costs 6 rolls on average — more than the first three faces together ($1+1.2+1.5 = 3.7$). With n equally likely faces (or coupons in cereal boxes) the answer is $n(1 + \frac{1}{2} + \dots + \frac{1}{n}) \approx n \ln n$; this is the **coupon collector’s problem**.

10.5 The hypergeometric distribution

Binomial sampling is *with* replacement (or from an inexhaustible population), so p never changes. What if we sample without replacement?

Definition 10.22 (Hypergeometric distribution). A population has N objects, of which M are “good” and $N - M$ are “bad”. Select n objects without replacement (all at once). Let $X :=$ the number of good objects selected. Then $X \sim \text{Hypergeometric}(n, M, N)$ with

$$\mathbb{P}\{X = x\} = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}},$$

valid for integers x with $0 \leq x \leq M$, $0 \leq n - x \leq N - M$ (and $n \leq N$).

The formula is pure counting: choose which x of the M good ones you got, choose which $n - x$ of the $N - M$ bad ones you got, and divide by the total number of samples.

Theorem 10.23. With $p := M/N$ (the population fraction of good objects),

$$\mathbb{E}[X] = n \cdot \frac{M}{N} = np, \quad \text{Var}(X) = \underbrace{\frac{N-n}{N-1}}_{\text{finite population correction}} \cdot np(1-p).$$

Remark 10.24 (Hypergeometric versus binomial). The mean is *identical* to the binomial mean. The variance carries an extra factor $\frac{N-n}{N-1} \leq 1$: sampling without replacement is more informative, so the count is less variable.

If $n \ll N$ then $\frac{N-n}{N-1} \approx 1$, and removing a few objects barely changes the composition of the population. In that regime

$$\mathbb{P}\{X_{\text{hyp}} = x\} \approx \mathbb{P}\{X_{\text{bin}} = x\},$$

which is why polling 1000 people out of 300 million is treated as a binomial problem. (Making this precise is a good exercise.)

Remark 10.25 (Where the mean comes from). Label the draws $1, \dots, n$. By symmetry the i -th object drawn is equally likely to be any of the N , so it is good with probability M/N — the same for every i , even though the draws are dependent. Linearity (Lecture 9) then gives $\mathbb{E}[X] = n \cdot \frac{M}{N}$ at once.

Example 10.26 (Where is the first ace?). A standard deck is shuffled. On average, in what position is the first ace? The last?

The four aces cut the other 48 cards into five gaps: before the first ace, between consecutive aces, and after the last. Fix one non-ace and look only at it and the four aces: all $5!$ orders of these five cards are equally likely, so the non-ace lands in each of the five gaps with probability $\frac{1}{5}$. By linearity, the expected number of non-aces before the first ace is $\frac{48}{5} = 9.6$, so the first ace is in expected position

$$9.6 + 1 = \boxed{10.6},$$

and by the same argument the last ace is in expected position $52 - 9.6 = \boxed{42.4}$. We never found the distribution of the first ace’s position — only a symmetry and linearity.